# InstallScout – technical reference

Version **1.8.74**. Short path: quickstart-en.md. Find the silent switches. The user workflow is in guide.md. This file covers engines, the command line, and Microsoft Graph. PDF: InstallScout-reference-en.pdf.

InstallScout does not run the installer. It reads the file, builds a silent command, wraps PSADT 4.1.8 as `.intunewin`, and creates a Win32 app.

## 1. Engines

An engine is recognized from byte markers in the file, PE section names, and sibling files. The highest score wins. `base_confidence` below is the starting value before extra evidence is added.

Wrappers (7-Zip SFX, WinRAR SFX, IExpress, Self Extractor, unknown EXE) are unpacked locally when the outer file has no reliable silent line. Clean engines such as Inno, NSIS, Wacom, and DDPM are not unpacked. InstallScout ships a bundled 7-Zip Extra `7za` for 7z payloads, so a system 7-Zip install is not required.

`/install` is not added. It is already the default for WiX Burn. `/norestart` is not added either. Inno Setup keeps its own `/NORESTART`.

`{file}` is the quoted installer path. `{product_code}` is the MSI `ProductCode`. `{app}` is the installed folder PSADT knows at uninstall time.

| Id | Name | Silent install | Alternative | Uninstall | Base |
|---|---|---|---|---|---|
| msi | Windows Installer (MSI) | msiexec /i "{file}" /qn | msiexec /i "{file}" /qb | msiexec /x "{product_code}" /qn | 0.97 |
| inno | Inno Setup | "{file}" /VERYSILENT /SUPPRESSMSGBOXES /NORESTART /SP- | "{file}" /SILENT /NORESTART | "{app}\unins000.exe" /VERYSILENT /NORESTART | 0.95 |
| wixburn | WiX Burn | "{file}" /quiet | "{file}" /passive | "{file}" /uninstall /quiet | 0.93 |
| ddpm | Dell Display and Peripheral Manager | "{file}" /Silent | "{file}" /Silent /CreateDebugLog="%TEMP%\DDPM-install.log" | "{file}" /uninst /Silent | 0.93 |
| advanced_installer | Advanced Installer | "{file}" /exenoui /qn | "{file}" /exenoui /passive /qn | "{file}" /exenoui /x // /qn | 0.92 |
| nsis | NSIS | "{file}" /S | | "{app}\uninstall.exe" /S | 0.90 |
| wacom | Wacom Tablet Driver | "{file}" /s | "{file}" /s /opt nowdc | "{file}" /s /u | 0.90 |
| msix | MSIX / AppX | Add-AppxPackage -Path "{file}" | Add-AppxProvisionedPackage -Online -PackagePath "{file}" -SkipLicense | | 0.90 |
| self_extractor | Self Extractor | harvested, otherwise the inner setup | | /uninst or /uninstall when present | 0.88 |
| bitrock | BitRock InstallBuilder | "{file}" --mode unattended --unattendedmodeui none | | | 0.88 |
| install4j | install4j | "{file}" -q | | | 0.85 |
| installaware | InstallAware | "{file}" /s | | | 0.80 |
| sevenzip_sfx | 7-Zip SFX | "{file}" -y | | | 0.80 |
| qt_installer | Qt Installer Framework | "{file}" --silent --accept-licenses | | | 0.80 |
| installshield | InstallShield | "{file}" /s /v"/qn" | "{file}" /s /sms | | 0.78 |
| winrar_sfx | WinRAR SFX | "{file}" /S | | | 0.75 |

| | | | | | |
|---|---|---|---|---|---|
| squirrel | Squirrel | "{file}" --silent | | | 0.72 |
| clickonce | ClickOnce | no reliable silent line | | | 0.70 |
| iexpress | IExpress | "{file}" /Q | | | 0.70 |
| wise | Wise Installer | "{file}" /s | | | 0.65 |

## Rules that differ from the template

- **NSIS.** /S must be capital. /D= must be last and unquoted.
- **Inno.** /S is not silent here. /NORESTART is Inno's own switch and stays.
- **InstallShield.** Newer wrappers pass /qn through /v"/qn". Older InstallScript may need a recorded setup.iss (/r, then /s /f1).
- **WiX Burn.** /quiet is the silent switch. /install is omitted because install is the default.
- **Advanced Installer.** /S and /VERYSILENT typically produce *Invalid command line*.
- **DDPM.** /Silent is not InstallShield /s /v"/qn".
- **Wacom.** The download is a 7-Zip SFX, but silent is /s on the EXE, not -y.
- **Self Extractor.** An outer /sAll, --silent, or /silent is kept. The Adobe-style chain becomes "file" /sAll /rs /rps /msi /quiet when /sAll or /rps is in the file. With no outer silent flag, the inner MSI/EXE is unpacked.
- **Squirrel.** Usually per-user. Context becomes User.
- **ClickOnce.** No silent template. Packaging for Intune is not a reliable target.
- **MSIX.** Detection uses the package name, not a product code.

## Suggested switches

At most 10. File flags come first (/analytics no, /sso no, and other name yes|no), then optional switches for the engine. /install and /norestart are not offered. A tick adds the switch to the silent line and invalidates the local .intunewin, so the package must be built again.

| Engine | Offered in addition to flags found in the file |
|---|---|
| MSI | ALLUSERS=1, REBOOT=ReallySuppress, /l*v "%TEMP%\install.log" |
| Inno | /NORESTARTAPPLICATIONS, /LOG="%TEMP%\install.log", /MERGETASKS=!desktopicon |
| NSIS | /NCRC |
| InstallShield | /f2"%TEMP%\install.log" |
| WiX Burn | /log "%TEMP%\install.log" |
| Advanced Installer | /exenoupdates |
| Wise | /l="%TEMP%\install.log" |
| DDPM | /TelemetryConsent=false, /HeadlessMode=true, /CreateDebugLog=%TEMP%\DDPM.log, /TurnOffCA |
| Wacom | /opt nowdc |

## Detection and the engine

Detection.ps1 writes to STDOUT and exits 0 when the app is installed. Empty STDOUT means not installed. The script does not throw.

| Source | Match |
|---|---|
| MSI ProductCode, or a GUID in the uninstall command | That uninstall key, and version >= |
| EXE without a product code | DisplayName, Publisher, and version >= |
| MSIX | Package name, and version when known |

A trailing Installer, Setup, Bootstrapper, or Driver word may be absent from DisplayName. Logi Options+ Installer matches Logi Options+, and Wacom Tablet Driver matches Wacom Tablet. The uninstall

key name counts too. A key named as the product still counts when DisplayName, publisher, and version are missing, as long as the key has some other value. A value that is present must still match. The publisher may differ when DisplayName is the product name plus a version and the name is not a short word. `WinRAR` matches `WinRAR 6.00 (64-bit)` even when the file says Alexander Roshal and the key says win.rar GmbH. `Git` still requires the right publisher. A leftover generic word such as Advanced is not used. Legal suffixes are ignored, so `Logitech, Inc.` matches `Logitech`. Versions are compared numerically. The same version and newer count. Older does not. A hyphen revision such as `6.4.14-1` is satisfied by both `6.4.14-1` and `6.4.14`.

Keys read: `HKLM` and `HKCU` under `SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall`, in both the 32-bit and 64-bit views. That still holds when the script itself runs as 32-bit.

## 2. CLI

`InstallScoutPortable.exe` with no arguments opens the GUI. A frozen EXE also opens the GUI unless a CLI flag is set. From source, `python -m installscout` behaves the same way.

Any of these flags forces the CLI even without `--cli`: `--json`, `--out`, `--psadt`, `--intune`, `--online`, `--sihq`, `--publish-intune`, `--switches`, `--icon`, `--supersede`, `--supersede-id`.

| Flag | Meaning |
|---|---|
| `paths` | Files or folders. Suffixes: `.exe`, `.msi`, `.com`, `.msix`, `.appx`, `.msixbundle`, `.appxbundle` |
| `--gui` | Open the GUI even when paths are present |
| `--cli` | Force text output |
| `--json` | Write the analysis as JSON on stdout |
| `--out PATH` | Write the same JSON to a file |
| `--psadt DIR` | Write the PSADT source, without `.intunewin` |
| `--intune DIR` | Write `.intunewin`, PSADT source, `Intune.txt`, and `Detection.ps1` |
| `--no-copy-installer` | Leave the installer in place. The package points at the original path |
| `--publish-intune` | Create the Win32 app. Requires `--intune` |
| `--icon PATH` | PNG, JPEG, GIF, ICO, or BMP for `largeIcon`. Without the flag, the icon comes from the EXE, the MSI, or a sibling file |
| `--supersede update\|replace` | Supersede older Win32 apps with a related name. At most 10 |
| `--supersede-id GUID` | Repeatable. Use known app IDs instead of a name search |
| `--tenant ID` | Empty means `organizations` |
| `--client-id ID` | Empty means Microsoft Graph PowerShell |
| `--online` | WinGet and Chocolatey. Only the product name and vendor are sent |
| `--sihq` | Open a name search on silentinstallhq.com. The command is not overwritten |
| `--switches TEXT` | Append arguments to the silent line for every file in the run |
| `--lang en\|da` | Language of CLI text. The GUI keeps its own choice |
| `-r` / `--recursive` | Search folders recursively. This is the default |
| `--no-recursive` | Only the given folder |
| `--version` | Print the version and exit |

Exit code `0` is success. Exit code `2` is used when no supported file was found, `--publish-intune` is missing `--intune`, or `--icon` cannot be read.

On startup, one anonymous call goes to Aptabase in the EU (`eu.aptabase.com`) with the version and language. File names and paths are not sent. `INSTALLSCOUT_USAGE=0` turns the call off.

```
.\InstallScoutPortable.exe --cli .\setup.exe
.\InstallScoutPortable.exe --cli .\setup.exe --online
.\InstallScoutPortable.exe --cli .\setup.exe --switches "/LOG=C:\logs\app.log" --json --out
```

```
.\analysis.json
.\InstallScoutPortable.exe --cli .\setup.exe --intune C:\out\intune
.\InstallScoutPortable.exe --cli .\setup.exe --intune C:\out\intune --publish-intune --icon
.\logo.png --supersede update
.\InstallScoutPortable.exe --cli .\setup.exe --intune C:\out\intune --publish-intune --
supersede replace --supersede-id <app-guid>
```

`--json` is a list of analysis objects: path, engine, confidence, commands, harvested switches, `option_flags`, VersionInfo, MSI properties, and notes. The first command with purpose `silent_install` is the recommended line.

The install line written to the portal is always PSADT, not the raw silent command:

```
.\Invoke-AppDeployToolkit.exe -DeploymentType Install -DeployMode Silent
.\Invoke-AppDeployToolkit.exe -DeploymentType Uninstall -DeployMode Silent
```

`setupFilePath` is `Invoke-AppDeployToolkit.exe`.

## 3. Graph

The Win32 fields `detectionRules`, `displayVersion`, and `maxRunTimeInMinutes` exist on Graph **beta**, not on v1.0. App calls therefore go to `https://graph.microsoft.com/beta`. `GET /me` for whoami goes to `https://graph.microsoft.com/v1.0`.

### Sign-in

OAuth 2.0 authorization code with PKCE. Device code is not used. A local HTTP listener on `http://localhost:<port>/` receives the code. The browser opens in the Edge work profile when that profile can be recognized.

| Authority | `https://login.microsoftonline.com/{tenant}/oauth2/v2.0/authorize` and `/token` |
|---|---|
| Default tenant | `organizations` |
| Default client | `14d82eec-204b-4c2f-b7e8-296a70dab67e` (Microsoft Graph PowerShell, public client) |
| Scopes | `DeviceManagementApps.ReadWrite.All`, `User.Read`, `offline_access`, `openid`, `profile` |
| Token file | `%LOCALAPPDATA%\InstallScout\graph.json`, Windows DPAPI for the current user (`ISC1`). Bound to the tenant and Client ID. Plaintext is deleted |
| Refresh | Refresh token. A 401 is retried once after refresh. An expired session deletes the token file. A different tenant or Client ID does not reuse the session |

A custom Client ID must be a public client with redirect `http://localhost` and the same permission. InstallScout does not create an app registration. The signed-in user must be able to create Win32 apps. The Entra role Intune Administrator, or the Intune role Application Manager, includes that. The tenant still needs admin consent for `DeviceManagementApps.ReadWrite.All`.

### Create and upload

1. `POST /deviceAppManagement/mobileApps` with `#microsoft.graph.win32LobApp`.
2. `POST .../microsoft.graph.win32LobApp/contentVersions`.
3. `POST .../contentVersions/{version}/files` with the encrypted size and the unencrypted size.
4. Poll the file until `azureStorageUri` is present.
5. `PUT` the encrypted blob in 6 MiB blocks.
6. `POST .../files/{id}/commit` with `fileEncryptionInfo`.
7. `PATCH` the app with `committedContentVersion`.

If upload or commit fails, the new app is deleted. Supersedence runs afterwards and does not delete the app if it fails.

Encryption matches Microsoft Win32 Content Prep: AES-CBC, a 256-bit key, HMAC-SHA256, `ProfileVersion1`, digest `SHA256`. The key, IV, MAC, and digest are sent at commit. The installer binary is not sent during the online lookup, only inside the encrypted `.intunewin`.

## App payload

| Field | Value |
|---|---|
| `displayName` | The name from the dialog, otherwise `{product} {version}`. At most 500 characters |
| `publisher` | Vendor, or the product name when the vendor is Unknown. At most 500 |
| `displayVersion` | Analyzed version. At most 50 characters |
| `description` | Engine, context, and detection summary. At most 4000 |
| `installCommandLine` / `uninstallCommandLine` | PSADT Silent, as above |
| `setupFilePath` | `Invoke-AppDeployToolkit.exe` |
| `applicableArchitectures` | `x64`, `x86`, or `arm64` |
| `minimumSupportedWindowsRelease` | `Windows10_22H2` |
| `installExperience.runAsAccount` | `system` or `user` |
| `installExperience.deviceRestartBehavior` | `suppress` (No specific action) |
| `installExperience.maxRunTimeInMinutes` | `60` |
| `allowAvailableUninstall` | `true` |
| `detectionRules` | One PowerShell script, `enforceSignatureCheck` false, `runAs32Bit` false, body base64 UTF-8 |
| `largeIcon` | Omitted when there is no logo. Otherwise `mimeContent` as PNG, JPEG, or GIF |
| `returnCodes` | `0` success, `1707` success, `3010` softReboot, `1641` hardReboot, `1618` retry |

The logo comes from an EXE/DLL/COM resource, from the MSI Icon table (ARPPRODUCTICON, otherwise a shortcut, otherwise the only icon), from `logo.png` / `icon.ico` next to the file, or from the file the user picks. Several icons with no ARPPRODUCTICON and no shortcut are left unused.

## Supersedence

Search is `GET /deviceAppManagement/mobileApps` with `$filter=startswith(displayName,'…')` and `$top=50`, at most 5 pages. The whole catalogue is not downloaded. The search term is the product name, not the edited Intune display name.

`POST /deviceAppManagement/mobileApps/{id}/updateRelationships` with `#microsoft.graph.mobileAppSupersedence`:

| supersedenceType | Meaning |
|---|---|
| `update` | Install the new app without uninstalling the old one first |
| `replace` | Uninstall the old app, then install the new one |

At most 10 targets. The GUI selects nothing automatically. The CLI with `--supersede` selects apps whose names look like the product.

Portal link after success:

```
https://intune.microsoft.com/#view/Microsoft_Intune_Apps/SettingsMenu/~/0/appId/{appId}
```

## 4. Windows Sandbox

**Try in Sandbox** is an optional GUI button. There is no CLI flag. It appears after **Analyze**, and only when `WindowsSandbox.exe` is installed. Packaging and upload stay available while the test runs. The result comes back in this window when the test finishes.

The button runs the current silent command inside a new Windows Sandbox, then the same `Detection.ps1` that upload sends to Intune. Empty detection output means not installed. The sandbox shows a message with the install exit code and that detection result. InstallScout reads the same text back into this window.

The command must contain the installer path. A package folder (the setup plus MSI, CAB, prerequisites, or a subfolder the installer needs) is mapped read-only at `C:\InstallScout\Installer`. A file directly in Downloads, Desktop, Documents, the OneDrive root, Temp, or the user profile is copied into the sandbox work folder, so the parent folder is not shared. That still applies when the folder has been redirected. A subfolder there, for example `Downloads\PSADT - CutePDF`, is still a package folder. The work folder is writable. Networking is enabled. On a PC with free RAM, the sandbox gets up to 8 GB of memory instead of the 4 GB default ceiling. Only one Windows Sandbox can run at a time. Closing the sandbox discards it. If you close it before a result comes back, the button turns back on so you can change the command and run it again. Nothing is installed on this PC.